

Bil Reference Manual

The documentation for Bil 2.13 (5 June 2026)
A modeling platform based on finite element/volume methods

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If you use Bil, we would appreciate that you mention it in your work. References, as well as the latest news about Bil development, are always available on <http://bil.ifsttar.fr>.

1 Introduction

Bil is an open-source software, based on finite element/volume methods, designed for the simulation of coupled phenomena in continuum mechanics, geomechanics, environmental engineering and materials sciences. Bil is distributed under the terms of the GNU General Public License (GnuGPL). The source code can be downloaded at

`http://bil.ifsttar.fr`

Bil is intended to be used by students, engineers or reseachers to work out problems or develop their own models. Bil is written in C++ language. Thus provided that a C++ compiler was installed, it can run on any OS like Windows, MacOS or Linux-based OS.

Bil is developped for 1D, 2D and 3D problems. It doesn't include a mesh generator and post-processing treatment of outputs. However it can read mesh files created by the open-source free software Gmsh downloadable at <http://www.geuz.org/gmsh/>. The output files created by Bil for 1D problems can be used easily by some plotting programs such as Gnuplot. But as a general rule, post-processing options create output files which can be used by Gmsh for post-processing treatments (see Bil options).

2 Running Bil

Bil can be run non-interactively only. To compute the solution of a problem described in the input data file `my_file`, type:

```
bil [options] my_file
```

The input data file format is described in the following section. Without any option, running Bil will create the output files `my_file.pi` and `my_file.ti` (see Chapter 4 [Output files], page 9). With the option `-h` (help) there is no input data file to give. The command

```
bil
```

(alone) displays the available options.

The way Bil will run may depend whether there are some other files (see Chapter 5 [Other files], page 11). The most important of them, namely `my_file.graph.iperm`, defines the way nodes are renumbered so as to optimize the storage of the LU decomposition of the matrix, in case of a Gauss elimination method (Crout's method). Except for 1D problems where nodes are numbered from one end to the other of the mesh, it is very important, if not essential, to create this file with an aquedate option, e.g. `-iperm`, (see Bil options). In case of multi-frontal methods (such as SuperLU or MA38) or Krylov space methods (such as those found in Petsc library), this file is not needed.

3 Input data file format

The file `my_file`, as mentionned above, provides the input data of the problem to be worked out. A list of reserved key-words (with an upper-case first letter) organizes the inputs in several groups such as mesh, material properties, boundary conditions, etc. For example the key-word **Geometry** is followed by the inputs defining the dimension and the symmetry of the problem. Any content of a line after `#` is considered as comments and skipped. The full list of reserved key-words which must appear in `my_file` is given in the table below.

key-word	description
Geometry	dimension and symmetry of the problem
Mesh	define the mesh
Material	material properties of the material index 1
...	...
Material	material properties of the material index n
Fields	define fields i.e. space functions
Initialization	define the initial conditions
Functions	define time functions
Boundary Conditions	define the boundary conditions
Loads	define the loads
Points	define some points for output files
Dates	define the dates for output files
Objective Variations	objective variations of the main unknowns
Iterative Process	convergence criteria of iterative process
Time Steps	time steps calculation

Some other optional input data can also be provided:

key-word	description
Units	change the units
Periodicities	periodicities of the mesh for peridiodic problems
Model	the names of unknowns can be changed

An on-line help is provided by typing:

```
bil -h
```


4 Output files format

Each run produces 2 sets of output files. In the first set, output files are named:

`my_file.pi`

where i is an integer ranging from 1 to the number of points defined by the key-word **Points**. If there is no point no file will be created. These files provide the results obtained at the specified points. The first column contains the times at which the results have been obtained. The other columns contain the value of specific quantities as implemented in the model defined in the key-word **Material**. In the second set, output files are named:

`my_file.ti`

where i is an integer ranging from 0 to the number of dates defined by the key-word **Dates**. These files provide the results obtained at the specified dates. The three first columns contain the three coordinates of nodes. The other column contain the value of the same quantities as those contained in the first set of files.

Some lines of these 2 sets of files are commented as indicated by the character **#** in the first column. These comments provide some informations about the nature of the computed quantities found in the following lines.

5 Other files

Bil produces some files and sometimes can read and use some other files. The name of these files are formed with the name of the input data file and suffixes, `my_file.suf`, in the same way as the output files. They are listed in the table below.

file	description
<code>my_file.ti</code>	output files related to date index i
<code>my_file.pi</code>	output files related to point index i
<code>my_file.posi</code>	view i to be read by Gmsh
<code>my_file.msh</code>	Gmsh mesh file
<code>my_file.graph</code>	mesh graph
<code>my_file.graph.iperm</code>	inverse permutations file
<code>my_file.sto</code>	storage file
<code>my_file.cont</code>	continuation file (see below for explanations)
<code>my_file.conti</code>	continuation file (see below for explanations)

The files `my_file.cont` and `my_file.conti` allow to continue a previous computation or to resume an interrupted calculation achieved with a previous input data file (`my_previous_file`). With `my_file.cont` the process doesn't go through the initialization stage (`ComputeInitialState`, see below) so that the calculation continues as if there hadn't been interruption. With `my_file.conti` the process goes through the initialization stage so that some variables of the model can be re-initialized (e.g. strain variables can be reset to zero). To do so, copy the file `my_previous_file.sto` in `my_file.cont` (or `my_file.conti`) and run bil with `my_file` as a new input data file in which you will have defined some additional dates beyond the last date defined in `my_previous_file`. Boundary conditions can also be modified in this new input data file in order to deal with changing from Dirichlet to Neumann boundary conditions.

6 Models

Mostly the concept of model refers to the constitutive equations or complementary laws that are needed to mathematically end up with a well-posed problem. However we need more informations, here, regarding the numerical methods that are used to handle complex problems and geometries. These methods are implemented in a single file whose basename identifies the code name of the model. Therefore the concept of model should be extended to fit with the set of informations pertaining to:

- The number and kind of equations to be solved
- The complementary laws that are needed to have a well-posed problem
- The informations associated to the numerical methods employed

Models aim at addressing the behaviour of the material at the scale of one finite element. Therefore the methods defined in object "Model_t" aim at computing matrix, residual forces, outputs and so on, for the nodes of one element. The object "Element_t" is therefore the main input entry of all methods of object "Model_t".

A short description of the available models can be displayed by typing:

```
bil -m
```


7 How to develop a new model?

To tell Bil to account for a new model, you just need to create a new file, e.g. `my_model.c`, in the folder `src/Models/ModelFiles` of the Bil's source directory and add the basename of this file, namely `my_model`, to the list of the available models found in `src/Models/ListOfModels.inc`¹. This model will be taken into account automatically the next time binary files will be created.

To help you in creating this new file, it is recommended to learn from already existing files. This file should contain at least the 11 methods of the Model class-like structure. These methods are listed in the table below.

method	description
<code>SetModelProp</code>	Set the model properties
<code>ReadMatProp</code>	Read the material properties
<code>PrintModelProp</code>	Print the model properties
<code>DefineElementProp</code>	Define some properties of the element.
<code>ComputeInitialState</code>	Compute the initial state
<code>ComputeExplicitTerms</code>	Compute the explicit terms
<code>ComputeMatrix</code>	Compute the matrix
<code>ComputeResidu</code>	Compute the residu
<code>ComputeLoads</code>	Compute the loads
<code>ComputeImplicitTerms</code>	Compute the implicit terms
<code>ComputeOutputs</code>	Compute the outputs

¹ Upper camel case convention is used to name the source files of this project. The same convention is used for the attributes and the methods of objects.

8 Examples

8.1 Drainage of a column

This problem is governed by the Richards' equation. A 1 meter high sand column is initially saturated. The liquid pressure is initialized as: $p_l = p_{atm} - g(x - 1)$. At $t = 0$ we drained the column from the bottom by imposing the pressure to $p_l = p_{atm}$. The input data file is given below.

inputs	comments
# Drainage of # a sand column	You can write some comments by beginning any line with #.
Geometry 1 Plan	Geometry of the problem 1D pb, plane symmetry
Mesh col.msh	Mesh The mesh is read in this file (format Gmsh). This mesh consists in a 20 elements mesh between 0 and 1. There are 2 regions. The region 1 is the point at 0. The region 2 is the line between 0 and 1. There is 1 material.
Material Model = M1 gravite = -9.81 phi = 0.3 rho_l = 1000 k_int = 4.4e-13 mu_l = 0.001 p_g = 100000 Curves = tab	Material 1 code name of the model gravity porosity fluid mass density intrinsic permeability fluid viscosity gas pressure in the file tab, there 3 columns: p_c S_l k_{rl}
Fields 2 Type = affine Value = 1.e5 Gradient = -9.81 Point = 1. Type = affine Value = 1.e5 Gradient = 0. Point = 0. Initialization 1 Region = 2 Unknown = p_l Field = 1	Fields 2 fields affine field defined by $10^5 - 9.81 * (x - 1)$ constant field equal to 10^5 Initial conditions 1 initial condition in the region 2, $p_l = 10^5 - 9.81 * (x - 1.)$.
Functions 0	Time functions $f(t)$ here there is no function
Boundary Conditions 1 Region = 1 Unknown = p_l Field = 2 Function = 0	Boundary conditions 1 boundary condition in the region 1, $p_l = f(t) * 10^5$ (by default $f(t) = 1$)

Loads

0

Loads

there is no load

Points

0

Points where we want outputs

no points

Dates

2

Dates where we want outputs

2 dates

0. 1800000

 $t_0 = 0$ and $t_1 = 1800000$

Objective Variations

 $p_l = 1000$

Objective variations of unknowns

objective variation $\Delta p_l = 1000$

Iterative Process

Iterations = 20

Parameters for the iterative process

20 iterations

Tolerance = $1e-10$ the tolerance is 10^{-10}

Repetitions = 0

no repetition

Time Steps

Dtini = 1

Parameters for time steps calculation

initial time step equal to 1.

Dtmax = 3600

maximum time step equal to 3600.

9 Version history

News in 2.10: Reorganization of the header files regarding the dependencies that only require to be forward declared and those that need to be #included.

News in 2.9: Use of the library "autodiff" via option at compiling time. Compiler c++ is forced for all source files (.c and .cpp). Introduction of the interface "MaterialPointMethod.h" for an easiest way of implementing models.

News in 2.8: Parallelization via OpenMP and MPI.

A new algorithm is implemented in the module SNIA which is based on a sequential non iterative approach (SNIA). The build system is switched to CMake. Bugs fixed in MA38 solver. New way of using extra libraries. The number of temporal discrete solutions kept in memory for possible use in time discretization schemes can be given in the command line. The solver KSP from the Petsc library can be selected if the external library "libpetsc.so" was installed (see EXTRALIBS). Re-arranging some allocation of memory space for the mesh. Reformulation of the jacobian matrix of the transformation between the element in the reference frame and that in the current frame. Zero-thickness element can be dealt with. Unknowns like pressure/temperature or other can be made continuous across such zero-thickness elements. The python script "utils/gmshcrack/gmshCrack.py" can be used to insert zero-thickness elements in mesh file generated by GMSH.

News in 2.7: Improvements on exception mechanisms, in no case results will be lost.

News in 2.6: The solver of sparse unsymmetric system of linear equations MA38 from HSL is introduced in the package and can be selected from the command line. The solver SuperLU can also be selected if the external library "libsuperlu.so" was installed (see the file "make.extralibs").

News in 2.5: A new interface "Session.h" is created. Methods like Session_Open and Session_Close can be used to open and close sessions. Sessions are in a stack with the current session at the top. Open a session = push it, close a session = pop and delete it. If needed instances like Message, Exception, etc... are created once in each session of the stack. Only those of the top session are currently used.

News in 2.4: Two new database interfaces: "HardenedCementChemistry.h" and "CementSolutionDiffusion.h" to be used for cement-based materials. New models now use these interfaces. The references used to implement some databases are now copied in the folder "src/Models/DataBases/References".

News in 2.3: Use of C++ compiler. Programmation is now extended to C++ language. Hence the use of .cpp and .hpp files is allowed. A new object "Exception.h" has been created to handle exception mechanisms such as interruption, floating point error. In such occurring event the program handles the event, saves outputs and exits in a clean way.

Introduction of the curve builder "Expressions", based on the evaluation of mathematical expressions obtained from AnaGram (www.parsifalsoft.com). In this version a shared library, libbil.so, from the genuine sources of Bil is created and installed in the machine. Linkage with other possible external libraries is possible. A new extension of input file ".conti" has been introduced so that we can load the solution from a previous calculation and continue this calculation while going through the initialization stage (see documentation).

News in 2.2: WARNING, the outputs of FVM_ComputeIsotropicConductionMatrix() and FVM_ComputeMassAndIsotropicConductionMatrix() have been modified see examples in models using these methods. New objects have been introduced: Views, TextFile, CurvesFile, MatrixStorageFormat. Introduction of the curve builder "Evaluate", based on the evaluation of mathematical expressions obtained from Snippets (www.brokersys.com/snippets/). Some bugs fixed, e.g. those in "Buffer_FreeFrom". Extended field delimiters in curves files. Some new models.

News in 2.1: A class-like structure Buffer_t has been introduced as a circular buffer. It can be used in any functions to compute vectors or tables of any type. Numerical Methods, such as FEM or FVM, have been implemented in files such as FEM.[c,h] and FVM.[c,h]. These methods are viewed as object with class-like structures FEM_t and FVM_t. In these files methods have been implemented such as FEM_ComputeMassMatrix and can be used directly in models provided that header has been included. The folders "Common" and "Main" have been created. We moved the main files in "Main" and files of common use, like "Buffer.c", in "Common". DataBases have been created which can be used in model files. The matrix storage format have been implemented in a separate file "MatrixStorageFormat.h".

News in 2.0: The code has been rebuilt to an object-oriented programming code. However the code is still implemented in C language. Class-like structures have been introduced and implemented in separated files. Each class-like structure has attributes and pointers to functions (see e.g. structure Model_t). Moreover some improvements have been introduced. The file names of models (in the folder "ModelFiles") can be chosen arbitrarily. The class-like structures "NodeSol_t" and "ElementSol_t" are two linked lists which contained the nodal and elemental solutions. A new pointer to "double" type in "ElementSol_t" can be used to store constant terms.

Nouveautés dans 1.8: création d'une structure "modl_t" contenant les méthodes (tâches élémentaires) i.e. des pointeurs sur fonctions.

Nouveautés dans 1.7: option "base" ajoutée dans le Makefile. Qqs nouveaux modèles. Option "Relative" possible dans OBJE, donnée après la valeur. Chargement automatique des modèles, il n'est plus besoin de modifier le fichier "xmod.c". Les fonctions "dm,qm,tb,ch,in,ex,ct,mx,rs,so" sont renommées "dmNB,qmNB,tbNB,chNB,inNB,exNB,ctNB,mxNB,rsNB,soNB". Les titres des modèles ainsi que des exemples de données sont gérés par les fonctions "qm1, qm2 ...". Options nouvelles de la ligne de commande. "bil -m" affiche les titres des modèles. "bil -m I" affiche

un exemple de donnees du modele I. Possibilite d'avoir des elements d'ordres multiples (1 pour chaque inconnue). Cette possibilite est geree par les tableaux "el.pin" et "el.peq". Une valeur negative associee a un noeud et une inconnue n'est pas prise en compte comme inconnue globale. Possibilite de creer de nouvelles fonctions d'interpolation au niveau des modeles dans "tb1, tb2, ..." avec la fonction "creer_interpolation(...)". Possibilite de definir des champs aux points d'intersection d'une grille dans l'espace avec l'option "Type = grille" dans CHMP (l'option Type = affine restant par default).

Nouveautes dans 1.6: bug corrige dans "sauvep". Les fichiers "mod.c" et "mod.h" sont renommés "xmod.c" et "xmod.h". Le specificateur de type "void" est remplace par "int" pour les fonctions "ex1(), ex2() ..." definies dans "m1.c, m2.c ..." et pour l'identificateur de type "ex_t" defini dans "defs.h". On utilise ce retour dans les fonctions "explicite()" et "algorithme()" definies dans "calc.c". Compatibilite avec le format de maillage version 2.0 de GMSH.

Nouveautes dans 1.5: definition des types de fonctions "dm_t", "qm_t", "tb_t", etc... Reorganisation du fichier "mod.c" par l'introduction de tableaux de pointeurs de fonctions pour simplifier l'introduction de nouveaux modeles. Possibilite de charger la librairie de SuperLU (voir le fichier "make.inc"). Creation de l'option de la ligne de commande "-m slu" qui permet d'utiliser la methode de resolution proposee par SuperLU a condition d'avoir construit, auparavant, le fichier des permutations inverses a l'aide d'un programme adapte comme Metis.

Nouveautes dans 1.4: reorganisation et creations des fichiers : calc.c, lecd.c, postt.c, renum.c. Suppression des variables statiques. Creation d'une structure de donnees "dnns_t", d'une structure pour la matrice "mtrx_t", d'une structure pour la solution "sltn_t", d'une structure pour la renumeration "nume_t". Amelioration de certaines fonctions de lib.c. Suppression de rssurf (pris en charge par rsmass).

Nouveautes dans 1.3: nouveaux modeles inclus. Qqs ameliorations apportees dans le calcul de dt (fonction pasdt). Apport de nouvelles info dans les structures mate_t, elem_t: en particulier neq,eqn,inc dans mate_t. Creation de la structure node_t et suppression du pointeur no_x. Cette version permet de prendre en compte des modeles reposant sur un nombre d'equations et sur des natures d'inconnues pouvant varier d'une region a l'autre. Cette version realise la continuite des equations en fonction de nouvelles informations contenues dans les modeles comme le nombre d'equations et les noms (predefinis) de chaque equation et inconnue associee. En consequence la structure du fichier a ete (legerement) modifiee pour prendre en compte ces informations, notamment dans les mots-cles INIT,COND,CHAR,ALGO (voir l'aide en ligne). Pour plus de clarte les donnees relatives aux variations objectives des parametres ont ete deplacees du mot-cle ALGO dans le nouveau mot-cle OBJE.

Nouveautes dans 1.2: creation d'un repertoire exemples. Simplification de la gestion des elements de surface (suppression de SURF). Creation de champs (mot-cles CHMP). Gestion des interruptions dans le calcul de la matrice.

Nouveautés dans 1.1: creation des pages info et de la doc sous differents formats (ps,pdf,txt). Creation des repertoires bin et lib. Amelioration de la procedure d'installation. Simplification du jeu de donnees par la definition des regions de maillage.

Appendix A License

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